Introduction to the Raft Protocol

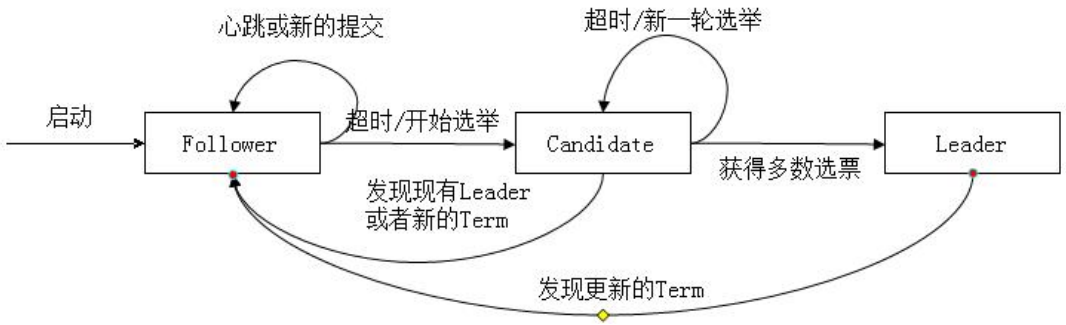
ghostwydwu(武勇栋)



System Architecture

Node States and Roles

- **Each node may assume one of three roles: **Follower**, **Candidate**, or **Leader**.**
- **- The Leader provides read-write services to external clients.**
- **- A Follower only acts as a request proxy.**
- **- A Candidate is merely an intermediate state and does not serve requests.**
- **A **Term** is the time interval starting from the beginning of one leader election until the start of the next election. Terms are represented by monotonically increasing integers.**



workflow

Commit Principle among Nodes

- The leader persists every modification operation using `LogEntry`.
- Commit requests in each term are executed strictly in the identical order.

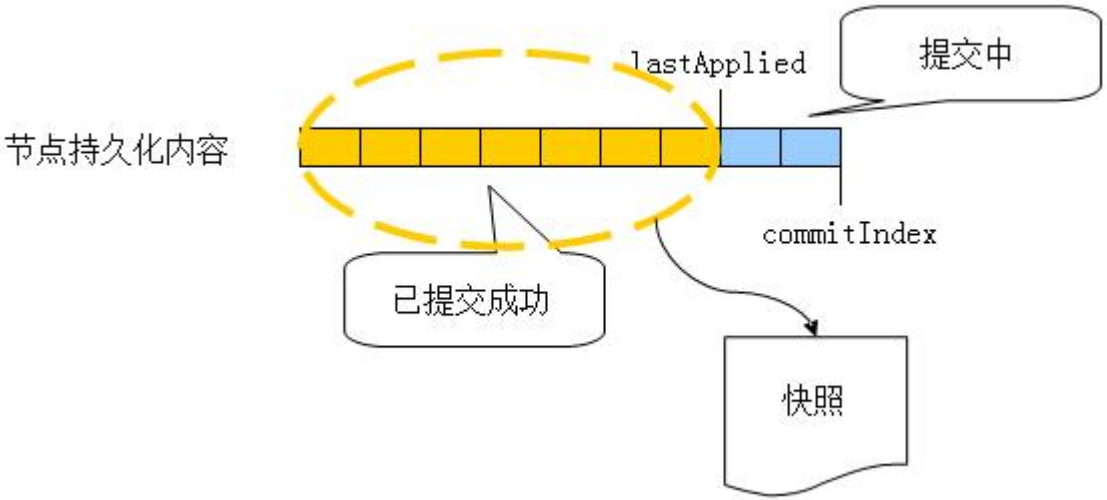
Distributed Commit

1. Each modification is assigned an incrementing sequence number as an **Entry**.
2. The leader first persists the Entry locally, then sends `AppendEntries` requests to all other nodes. The request carries the Term number, Entry content, and the index of the last Entry already applied to the Replicated State Machine (RSM).
3. Each follower validates continuity of its local entries, appends the incoming Entry locally, and returns a success response.
4. After the leader receives successful responses from a majority of nodes, it marks this Entry as committed and merges it into its own state machine.
5. Every `AppendEntries` request carries the index of the highest successfully-committed entry known to the leader. Upon receipt, each Follower persists the entry and updates its local state machine accordingly.
6. When requests and responses exceed the heartbeat interval, the leader proactively sends empty `AppendEntries` requests as heartbeats, advertising the index of the most recently committed entry.

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Log Compation

- **Log Compaction**
  - **Each node independently generates snapshots of its state machine, triggered by timer thresholds and log-entry count limits. All log entries predating the snapshot point are then deleted.**
  - **Since metadata for CacheMaster is relatively small, **full snapshots** are adopted here.**
  - **Full snapshots are managed in a rolling fashion, retaining a fixed number of snapshot files.**



Fault Detection And Election

- Fault Detection
- A Follower considers the leader faulty if it receives neither an `AppendEntries` request nor a heartbeat within the timeout window (timeout configured as 3 s = 1 s × 3). A heartbeat timeout triggers leader - failure detection.

- Election Process
- After heartbeat timeout, a Follower transitions to the Candidate state and initiates an election. It broadcasts `RequestVote` requests containing its current Term number and the index of its latest log entry.

- Rules for receiving nodes:
- - If the receiver’ s local Term number is larger, reject the vote request.
- - If the receiver has not yet cast its vote or has already voted for this candidate, **and** the log entry in the incoming request is at least as up - to - date as its own log, grant the vote.

- Once a Candidate receives votes from a majority of nodes, it becomes the Leader and immediately sends an empty `AppendEntries` as a heartbeat.
- Candidates use randomized election timeouts to avoid conflicts during leader elections.

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Fault Recovery

- After a successful election, the new Leader broadcasts its status by sending an empty `AppendEntries` message carrying its Term and entry index.
- Upon receiving the leader’ s heartbeat broadcast, all other Followers and Candidates switch to the Follower state and update their local Term and leader identifier.

- - If the entry index in the initial empty heartbeat matches the local log continuity, the Follower returns success directly.
- - If log continuity cannot be matched, the Follower returns failure and reports its highest local entry index to the leader.

- When the leader receives a failure response from a Follower：

- 1. If the leader still holds the missing log entries, it sends all missing entries via `AppendEntries`.
- 2. If those missing log entries have already been deleted on the leader, the leader sends a full snapshot of its state machine, and then proceeds to synchronize the remaining missing log entries.

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Cluster-Wide Crash and Recovery

- If the entire cluster loses its data, cluster metadata can be restored from CacheNode instances:
  - Set - level information can be exported and restored from configuration files of the logic module.
  - CacheNode metadata can be exported and restored from the OSS database.
  - Block metadata and status information can be pulled and recovered from each CacheNode.
  - Mapping relationships between Blocks and stripes are persisted within each Block. In case of duplicate mappings, the record with the larger `fid` takes precedence.

- After Block recovery
  - Check the maximum `fid` within the Set.
  - If maximum fids are consistent, set the Block state to `FREE`. Otherwise set it to `MOVING`.
  - Resume write operations once synchronization completes.

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Reliability Estimation

Reliability Assumptions

- - Class - C machine monthly crash probability: 1%. Assume repair completes within one day.

No.	shard number	Annual Failure Rate
1	3	1.2E-04
2	5	1.3E-07

- - Class - B machine monthly crash probability: 0.4%.

No.	shard number	Annual Failure Rate
1	3	1.9E-05
2	5	8.3E-09

- Deployment Guidance
  - For Class - B machines: a 3 - node deployment suffices.
  - For Class - C machines: a 5 - node deployment is recommended.
  - Virtual machines with rack - aware disaster tolerance are also supported.

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Thanks!

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